

Project 3

The Books Don't Match

Professional Reconciliation Report

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AI Tool Used: ChatGPT

Dataset: Farewell Gift Reconciliation Example

Core finding: The raw payment record is PKR 1,500 above the expected total, but PKR 3,500 is still outstanding from named contributors. The apparent surplus is explained by a duplicate payment and an unidentified transfer.

1. Executive Summary

This project reconciles an 18-person farewell-gift roster against a messy digital payment export. It applies personal interpretation rules to inconsistent names and memos, separates valid payments from exceptions, and produces a named follow-up list.

2. Problem Statement

A bank or wallet total can appear correct while individual obligations remain unpaid. The objective was to compare a known hand-counted total of PKR 45,000 against the raw digital record and determine who paid, who still owes money, what should be refunded, and which transfer cannot yet be identified.

3. Dataset and Ground Truth

Item	Value
Confirmed contributors	18
Fixed share per contributor	PKR 2,500
Expected total	PKR 45,000
Raw digital total	PKR 46,500

4. Interpretation Rules

The solution used only the collector's documented rules: unique nicknames and short forms were mapped to roster names; Umer's combined payment was split; the spouse payment was credited to Osman; Hassan's repeated transfer was flagged as a duplicate; Mahnoor's payment remained partial; and the unregistered transfer was not allocated.

5. AI Workflow and Prompt Iterations

The initial prompt requested a reconciliation script. Later prompts added strict identity rules, separate exception reports, a raw-total control, and a requirement to name each person and exact amount requiring action. The complete prompts are included in prompts.md.

6. Reconciliation Results

Measure	Amount
Expected roster total	PKR 45,000
Raw digital payment total	PKR 46,500
Valid credits to roster	PKR 41,500
Duplicate transfer	PKR 2,500
Unidentified transfer	PKR 2,500
Outstanding roster dues	PKR 3,500
Control difference	PKR 0

7. Follow-Up Register

Person / Item	Finding	Required Action
Mahnoor Iqbal	PKR 1,500 credited; PKR 1,000 short	Collect PKR 1,000
Faisal Mehmood	No matched transfer	Collect PKR 2,500
Hassan Sheikh	Second PKR 2,500 transfer is duplicate	Refund or credit
Unregistered sender	PKR 2,500 cannot be allocated	Identify manually

8. Manual Verification

The raw record was independently checked: 15 ordinary PKR 2,500 transfers equal PKR 37,500; Umer's combined transfer is PKR 5,000; Mahnoor paid PKR 1,500; and the unregistered transfer is PKR 2,500. The total is PKR 46,500. The program's control also reconciles PKR 41,500 valid credits + PKR 2,500 duplicate + PKR 2,500 unidentified = PKR 46,500, leaving a zero difference.

9. What Worked

The script handled inconsistent capitalization, initials, nicknames, spouse payments, combined payments, duplicates, partial payments, ambiguous notes, and unidentified senders. It produced both a control total and a practical follow-up list.

10. Challenges and Limitations

The unidentified transfer cannot be assigned without external confirmation. This is a necessary control rather than a failure. The workbook also identifies the dataset as a fictional, illustrative example, so the report should be presented as a demonstration.

11. Conclusion and Learning

A total that looks sufficient can conceal unpaid individuals. The key learning is that reconciliation must follow identity, purpose, and exception rules - not merely compare two headline totals. AI was useful for automating the rules, but the zero-difference control and named follow-up register made the result trustworthy and actionable.